

QUIZ

Roll no: SE-24058

Question 1:

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As a senior lead at Netsale I do not agree with the management's decision of not hiring a software architect. The reason is that to deliver a quality software product careful planning of the system is required before developing it. This can be achieved by simplifying the software requirements which can be done by visualizing the system and design the implementation which would speed up the development process. A software architect performs these tasks.

The developers cannot fulfill these duties as they will be more concerned about coding the entire system rather than analyzing the system from different perspectives and considering all the system constraints. Hence, a software architect is needed to deliver a quality software system.

~~For example: A client asks Netsale~~

For example: A client asks Netsale to deliver an e-commerce application for his business. Now the deadline is tight as well as the requirements clearly reflect that the delivered system should be able to handle loads efficiently with along with secure transactions management. Now since the time is short the team without a software architect would make designs quickly for visualizing the system but there are high chances that they would miss something from the requirements or the proposed architecture was not considered carefully which can lead to faulty application. This scenario explains the need of hiring a software architect.

IT Environment

IT environment is one of the main components of software architecture that we need to take care of. IT environment ensures that the software system is developed, keeping in mind the hardware, network, cloud infrastructure and other resources in mind. For example there is a project of developing a Payroll system for a company which will be used by the HR department of that company. Since the software would be used within a company so it can be connected through LAN. So we need to make sure that the software is designed keeping this IT factor in mind.

QUESTION 2:

Client-Server Architecture: The client-server architecture, also known as request-response is an architecture style which operates on the principle of server handling multiple client requests at a time. It initiates when a client sends a request & ends with a response from the server. The main characteristics are:

- The system is centralized which means it allows resource sharing b/w client and server and also between client and client.
- The server can handle multiple clients at a time.
- The client & server interact with each other through a network (LAN, WAN, Internet etc.).
- It is a scalable architecture as multiple clients can be added without affecting the whole system.

1. Gmail: Google's Gmail is an email service provider which operates on CS architecture. Multiple clients (users) can send and receive emails (resource sharing) while the server manages it all.

2. Chrome: The chrome web server which allows internet surfing. It works on the CS architecture. The user (client) searches something and the server returns it.

results in the form of multiple webpages (response) if the connection is successful b/w the client (user's machine) and server.

QUESTION 3:

Technical Requirements:

- The system will allow the citizens to dispose the garbage through IoT enabled smart bins which will automatically detect the nearby dumped garbage.
- The system will notify the municipal authorities about the localities which have high dumped wastes by pop-up notifications.
- The system will allow municipal authorities to monitor and analyze the garbage and waste dumped through proper everyday analysis using pie charts, bar charts and graphical trends.
- The system will alert the registered recycling companies to recycle the waste they can utilize.
- The recycling companies can buy & waste for recycling through online payment processing including service & platform fee and will also get reward points after the

Non-Technical Requirements

- The system will handle all the transactions securely.
- The system will be able to efficiently & instantly detect the nearby garbage.
- The system should be scalable enough to handle multiple users (citizens, authorities, companies).

